

P05: Stardew Valley

Team Name: DeadDuckies

Roster: Emaan Asif (PM), James Lei (Dev 1), Sarah Zou (Dev 2), Kiran Soemardjo (Dev 3),

TARGET SHIP DATE: 2026-06-08

Site Location: <https://stardewvalley.app/>

Description

We are creating a Stardew Valley emulator with our initial phase focused on the standard farm map.

Games can be joined by players through the lobby in a pregenerated map. Players will have inventories, be able to sell and buy goods, fish, mine, craft, and establish relationships.

Map / Tiles / Movement: The world consists of a scrolling, tile-based map divided into regions (farm, town, forest, mines). Tiles represent terrain like water and trees. Players will be able to move across the map tile to tile.

Inventory: Players will have an inventory that stores all collected / bought items, including crops, materials, tools, and crafted goods. Items can be consumed, sold, or used for crafting and upgrades.

Farming: Crops and materials are produced through player actions and time progression on the farm.

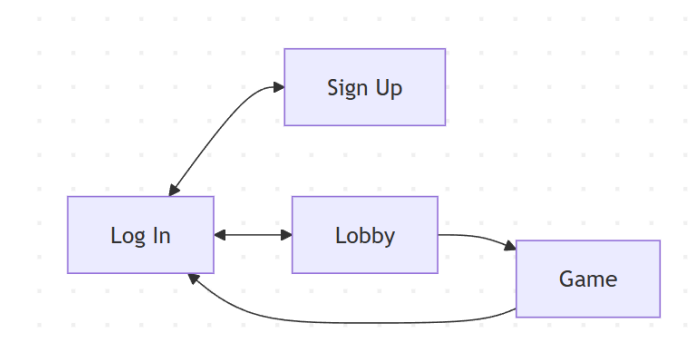
Time / Stamina: The game runs on a daily cycle: time will affect NPCs and activities. Through actions, players will lose stamina.

NPCs / Relationships: NPCs function as static sprites for buying and selling goods. Relationships can also be established with these NPCs.

Mining / Fishing / Foraging Systems: These activities allow players to collect resources, and are specific to their maps.

Crafting / Tool upgrades: Players can convert materials into functional items (e.g., chests, furnaces) through crafting. Tools can be upgraded to improve efficiency with mining / fishing / foraging.

Site Map



Log In / Sign Up - Users create accounts and log in

- Lobby
 - Users create a game or join other users' game
- Game
 - Displays the game

External Resources

Web XNB Converter - Extract Game Files

(<https://lybell-art.github.io/xnb-js/>)

Dependencies

Framework: None

JavaScript: SocketIO (<https://cdn.socket.io/4.8.1/socket.io.min.js>)

Python: Flask, Sqlite3, Flask-SocketIO

APIs: None

Tasks + Assignments

Project Manager: Emaan Asif

- (Phase 1) Inventory
- (Phase 2) Pierre's
- (Phase 4) Crafting
- (Phase 4) Tool Upgrades
- (Phase 5) Recipes

Dev1: James Lei

- (Phase 1) Entities
- (Phase 1) Player
- (Phase 2) Travel
- (Phase 3) Trees
- (Phase 4) Houses
- (Phase 4) Static NPCs V2 (Gus, Clint)

Dev2: Sarah Zou

- (Phase 1) Farm Map
- (Phase 2) Town Map
- (Phase 2) Stamina
- (Phase 3) Forest / Foraging
- (Phase 3) Mining Map
- (Phase 4) Eating

Dev3: Kiran Soemardjo

- (Phase 1) Time
- (Phase 2) Static NPCs V1 (Willy, Pierre)
- (Phase 2) Relationships
- (Phase 3) Fishing
- (Phase 4) Static NPCs V3 (Sebastian, Sam, Penny, Haley)
- (Phase 5) Community Center

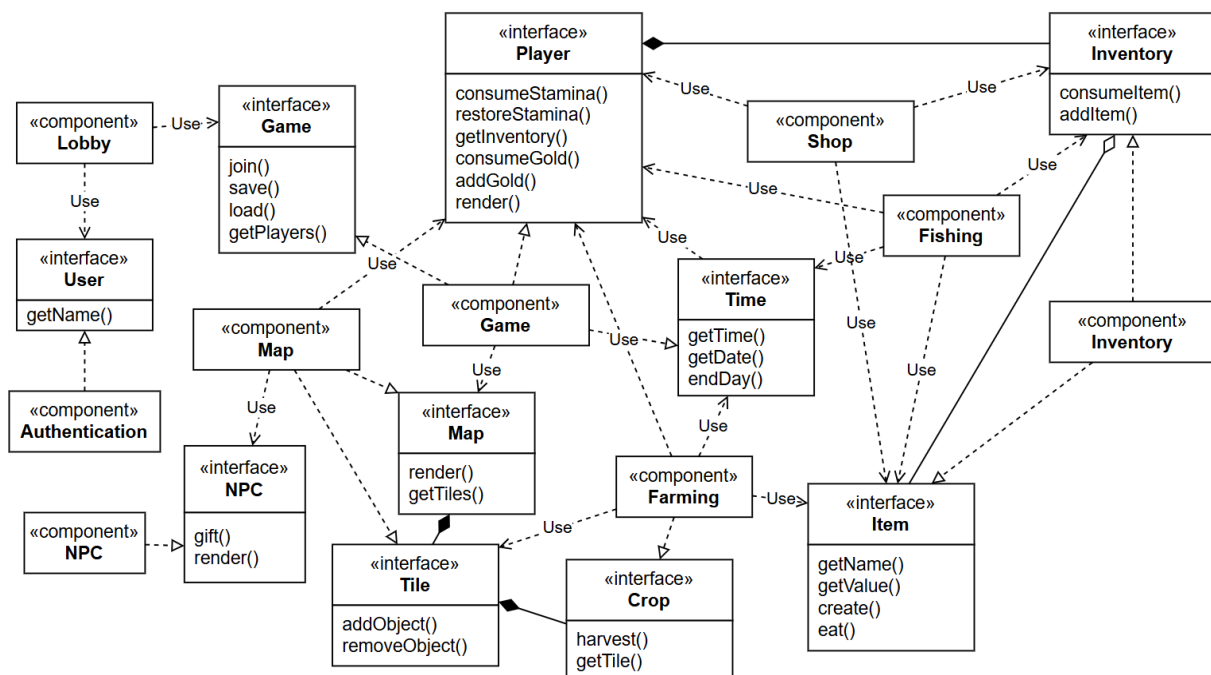
Extension Goals (Unassigned)

- Skill Levels
- Sound effects/music

Components

Authentication	- Allows users to create accounts and log in
Lobby	- Displays available games and let users create, join, and load games - Uses the User interface to associate games with users
Game	- Synchronizes games across players and saves game states - Uses the Map interface to associate a map with a game
Map	- Renders the map, the player, and all NPCs - Allows the player to travel between different areas and move
Farming	- Allows the player to raise crops - Uses the Player interface to consume stamina - Uses the Time interface to reflect crop growth time - Uses the Item interface to create an item upon harvest
Fishing	- Uses the Inventory interface to give a fish to a player - Uses the Item interface to create a new fish - Uses the Time interface to determine available fish species - Uses the Player interface to consume stamina
Shop	- Allows the player to sell and purchase items - Uses the Item interface to obtain the gold equivalent of an item - Uses the Player interface to give or take gold from a player - Uses the Inventory interface to give or take an item from a player
Inventory	- Allows the player to store and craft items
NPC	- Creates NPCs for the player to interact with

Component Map



Database Organization

Players		
TEXT	name	UNIQUE
TEXT	password	

Games		
INTEGER	id	PK AUTOINCREMENT
TEXT	player1	FK REF players(name)
TEXT	player2	FK REF players(name)
TEXT	player3	FK REF players(name)
TEXT	player4	FK REF players(name)
INTEGER	day	DEFAULT 0
INTEGER	money	DEFAULT 500

Tiles		
INTEGER	game_id	FK REF games(id)
TEXT	map	
INTEGER	x_pos	
INTEGER	y_pos	
TEXT	furniture	DEFAULT “[]”
BOOLEAN	tiled	

Inventory		
INTEGER	game	FK REF games(id)
TEXT	player	FK REF player(username)
TEXT	contents	
INTEGER	upgrades	DEFAULT 0

Plants		
INTEGER	game_id	FK REF games(id)
TEXT	map	
INTEGER	x_pos	
INTEGER	y_pos	
TEXT	type	
INTEGER	age	DEFAULT 0

Placeables		
INTEGER	game_id	FK REF games(id)
TEXT	map	
INTEGER	x_pos	
INTEGER	y_pos	
TEXT	type	
INTEGER	item	
INTEGER	progress	

Relationships		
INTEGER	game	FK REF games(id)
TEXT	npc_name	
TEXT	player	FK REF player(username)
INTEGER	points	
INTEGER	giftcount	DEFAULT 0
TEXT	status	